

Multiple Choice

1. **Answer:** B

Options: A) names for the layers of gaseous and metallic hydrogen deep within the planet.; B) alternating bands of rising and falling air at different latitudes.; C) alternating regions of charged particles in Jupiter's magnetic field.; D) regions of the plasma torus created by ions from Io's volcanoes

Note: Correct option: B - alternating bands of rising and falling air at different latitudes.

2. **Answer:** B

Options: A) The resonance pulls Io in different directions and generates heat.; B) It makes the orbit of Io slightly elliptical.; C) It creates a gap with no asteroids between the orbits.; D) It prevents formation of the ring material into other moons.

Note: Correct option: B - It makes the orbit of Io slightly elliptical.

3. **Answer:** D

Options: A) Triton and Desmos; B) Triton and Deimos; C) Phobos and Triton; D) Phobos and Deimos

Note: Correct option: D - Phobos and Deimos

4. **Answer:** A

Options: A) From the dismantling of small moons by impacts; B) From fragments of planets ejected by impacts; C) From dust grains that escape from passing comets; D) From accretion within the solar nebula at the same time the planets formed

Note: Correct option: A - From the dismantling of small moons by impacts

5. **Answer:** A

Options: A) Its rotation axis is nearly perpendicular to the plane of the Solar System.; B) It does not have an ozone layer.; C) It does not rotate fast enough.; D) It is too close to the Sun.

Note: Correct option: A - Its rotation axis is nearly perpendicular to the plane of the Solar System.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Hawking radiation. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** Orion. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.